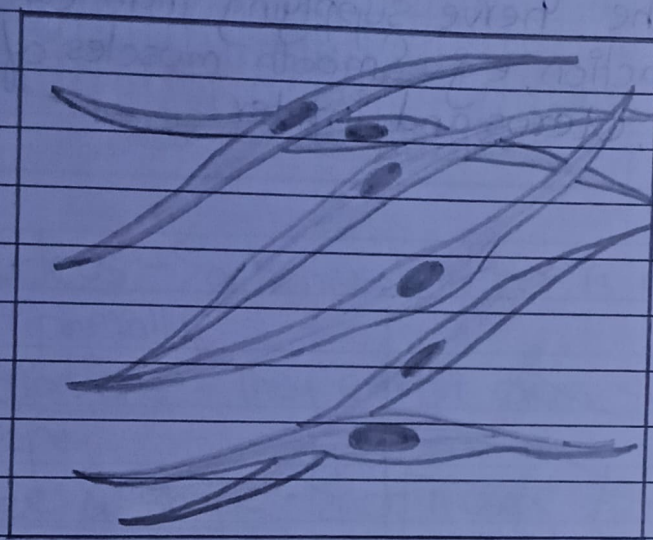


Muscles Tissue

Muscle is a contractile tissue, which brings movement of the body.

Types:- There are three types - skeletal, cardiac and smooth muscles.

1. Smooth muscles



→ Nucleus:- Each muscle fiber contains a single nucleus placed centrally.

Striations:- They do not exhibit cross striations and are smooth in form, supplied by autonomic nerves, hence involuntary.

Size & shape:- Each muscle fiber is an elongated spindle shaped cell. The length of the smooth muscle is highly variable (15-500 μm).

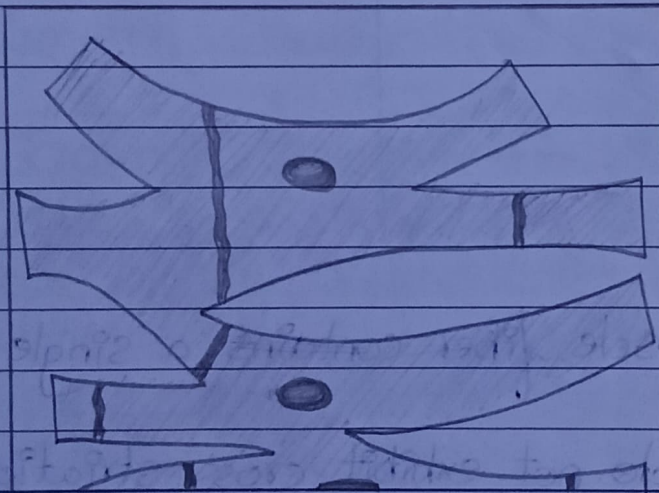
Location:- They often aggregate to form bundles and fascicles. They are found in the walls of gastrointestinal tract, respiratory tract, urogenital tract, blood vessels and a few muscles of the eye. They are arranged circularly inside and longitudinally outside in the walls of the gastrointestinal tract, urogenital tract, etc.

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Subtypes :- a. Multi-unit smooth muscles: Nerve fibers establish direct contact with several myocytes. Muscles contract under nervous stimulus, eg. Smooth muscles of iris and large arteries.

b. Unitary Smooth muscles: They have their own rhythmic contractility that is independent of nerve supply. Contraction of muscle is stimulated by stretch. However, the nerve supplying them can alter the rate of contraction, e.g. Smooth muscles of the stomach, intestine, uterus and ureter.

2. Cardiac muscle



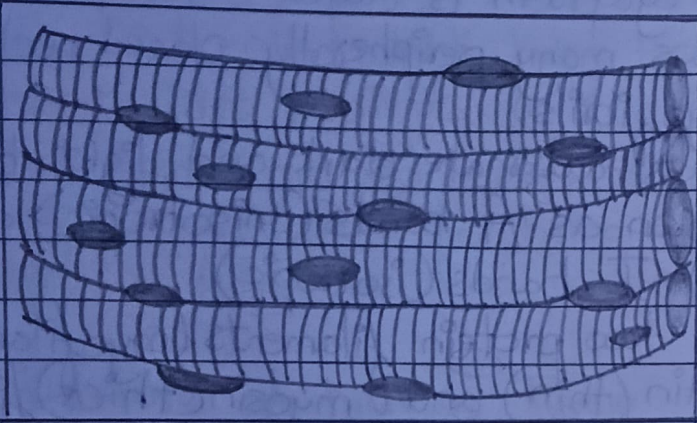
→ Nucleus:- Each muscle fiber has a single rounded nucleus placed centrally.

Striations:- It forms the myocardium of the heart, shows striations but is involuntary. It is meant for automatic and rhythmic contractions.

Size & Shape:- Myocytes are about $80\text{ }\mu\text{m}$ and $15\text{ }\mu\text{m}$ broad. Each muscle fiber has a single branches and anastomoses with the neighbouring fibers at "intercalated discs".

Location:- It forms the myocardium of the heart.

3. Skeletal muscle.



→ Nucleus:- Each muscle fiber is multinucleated. placed peripherally.

Striations:- They exhibit cross striations under the microscope.

Size & shape:- Each muscle fiber is a cylindrical cell containing a group of muscle fibrils.

Location:- These are most abundant, found attached to the skeletal system. like muscles of the limb and body wall. They are supplied by somatic (cerebrospinal) nerves, hence under voluntary control.

Blood Supply.

- The blood vessels and nerves enter the muscle at a point called neurovascular hilum.
- The arteries branch repeatedly to form arteriole and capillaries in the muscle.

Microscopic Structure of the Skeletal Muscle.

- Each muscle is made up of a number of muscle fibers.
- Each muscle fiber is multinucleated and shows striations.

They are also called myocytes and their length varies from 1-300 mm.

- The outer membrane of the muscle fiber is called sarcolemma and its cytoplasm is called sarcoplasm.
- The sarcoplasm shows many peripherally placed nuclei and large number of myofibrils.
- Each myofibril shows alternate dark and light bands.
- Dark bands are known as A bands (anisotropic) and the light bands and I bands (isotropic).
- Each myofibril has two protein filaments (myofilaments).
- They are called actin (thin) and myosin (thick) filaments.
- The dark band (A band) is the area where the actin and myosin filaments overlap each other.
- Only actin filaments occupy the light band (I band).
- This arrangement of actin and myosin filament is responsible for striations of skeletal muscle.
- In the middle of the dark band there is a light H band where there are no actin filaments.
- In the middle of the I band there is a dark Z line.
- The part of the myofibril between two Z lines is called sarcomere.

Nerve Supply.

The nerve supplying the muscle is called motor nerve.

Function of Nervous Tissue.

- Neurons generate and carry out nerve impulses. They produce electrical signals that are transmitted across distances, they do so by secreting chemical neurotransmitters.
- Responds to stimuli.
- Carries out communication and integration.
- Provides electrical insulation to nerve cells and remove debris.
- Carries messages from other neurons to the cell body.

Nervous Tissue.

- Nervous or the nerve tissue is the main tissue of our nervous system.
- It monitors and regulates the functions of the body. Nervous tissue consists of two cells: Nerve cells or neurons and glial cells. which helps transmit nerve impulses and also provides nutrients to neurons.
- Brain, Spinal cord and nerves are composed of nervous tissue, they are specialized for being stimulated to transmit stimulus from one to another part of the body rapidly.

Structure of Nervous Tissue.

- It is made of nerve cells or neurons, all of which consists of an axon. Axons are long stem-like projections emerging out of the cell, responsible for communicating with other cells called the Target cells, thereby passing impulses.
- The main part is the cell body which contains the nucleus, cytoplasm and cell organelles. Extensions of the cell membrane are referred to as processes.
- Dendrite is a highly branched processes, responsible for receiving information from other neurons and synapses (specialized point of contact). Information of other neurons is provided by dendrites to connect with its cell body.
- Information in a neuron is unidirectional as it passes through neurons from dendrites, across the cell body down the axon.

Nervous Tissue Location.

The nerve tissue or the nervous tissue is the chief tissue component of the two major parts of the nervous tissue - Central nervous system (CNS) formed by the spinal cord and the brain and the branching peripheral nerves of the peripheral nervous system (PNS) that control and regulate the functions of their activities.

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